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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Traiblazers
Project / Solution Name	SkillTide
Name (Leader)	HUANG BAONING
Email & Phone (Leader)	SWE2609018@xmu.edu.my · +60 1123419021
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://skilltide.replit.app (Replit, main deployment) Mirror: https://xmumhkt.arcueidmp.com (Cloudflare)
Code Repository URL (public / access granted to judges)	https://github.com/ArcueidMP/skilltide (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

SkillTide is one TypeScript code base: the same scoring engine runs on the server and, as a fallback, in the browser (diagram in Appendix A).

- **Data:** a Python pipeline (standard library only) downloads public data (O*NET, Anthropic Economic Index, Indeed and more; see 4.2) and builds one versioned JSON dataset, published only if its tests pass. There is no database.
- **Front end:** a Next.js static site in English, Malay and Chinese. PDF.js reads a CV inside the browser; the profile and progress stay in browser storage.
- **API:** /api/v1 (outlooks, matches, scenarios, cohort, Guide) runs as a Node.js server on Replit Autoscale (skilltide.replit.app) and as a Cloudflare Worker (xmumhkt.arcueidmp.com). Requests carry skill levels, goals, hours and budget, never the CV. If the API is unreachable, the browser computes the same results.
- **AI:** the Guide answers with fixed rules. When a student turns on AI wording, the server sends the question and SkillTide's computed facts to OpenAI's Responses API (gpt-6-luna) and shows the reply only if every number in it appears in those facts.
- **Delivery:** GitHub Actions tests, lints and builds every push to main; Cloudflare Workers Builds deploys main; the Repl pulls from GitHub and is republished.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Main language: interface, API, scoring engine, scripts	Apache-2.0
Python 3.13 (standard library only)	Data pipeline that builds the dataset	PSF License
Node.js 24 (CI) and 22 (Replit)	Builds, tests, and the API server on Replit	MIT

Name & Version	Purpose in Project	License
Next.js 16.4.2	Web framework; static export of the site	MIT
React 19.3.0, React DOM 19.3.0	User interface	MIT
Tailwind CSS 4.3.3, tailwind-merge 3.7.0, clsx 2.1.1	Styling	MIT
Recharts 3.10.1	Skills radar and trend charts	MIT
Motion 14.4.0	Interface animation	MIT
Lucide icons (lucide-react 1.60.0)	Icons	ISC
Mozilla PDF.js (pdfjs-dist 6.4.299)	Reads CV text in the browser	Apache-2.0
Source Serif 4 5.3.1, IBM Plex Sans 5.3.0, IBM Plex Mono 5.3.0 (Fontsource)	Self-hosted web fonts	SIL OFL 1.1
Vitest 5.0.5, tsx 4.23.15, ESLint 9.39.5, eslint-config-next 16.4.2	Tests, scripts and linting	MIT
Wrangler 4.151.3 (Cloudflare)	Builds and deploys the Cloudflare Worker	MIT or Apache-2.0
Demo film: three.js 0.186.1; Playwright 1.64.1 (playwright-core) driving Chrome	Draws the film's chart scenes; captures the app's screens	MIT; Apache-2.0
Demo film: FFmpeg 8.0.1 (gyan.dev full build)	Encodes the video, cleans and mixes the audio, runs the Whisper check	GPL-3.0 (this build)
Demo film: kokoro-onnx 0.6.1, ONNX Runtime 1.31.0, NumPy 2.5.4, SciPy 1.18.1, soundfile 0.14.0	Narration synthesis for an earlier cut, music, sound effects and the narration timing	MIT; MIT; BSD-3-Clause; BSD-3-Clause; BSD-3-Clause

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI Responses API (OpenAI), model gpt-6-luna	Optional "AI wording" for the Guide's answers, off by default; called only by our server, at run time	Paid: the team's own API key
UN SDG API v5 (UN Statistics Division)	SDG goal and target wording, read when the dataset is built	Free, public
Microsoft Learn Catalog API (Microsoft)	Free learning paths and their durations, read when the dataset is built	Free, public
HN Search API (Algolia, for Hacker News)	Counts skill mentions in monthly "Who is hiring?" threads, when the dataset is built	Free, public
Hugging Face Hub (Hugging Face)	Downloads the Anthropic Economic Index files when the dataset is built	Free, public
GitHub file downloads (GitHub)	Downloads the Indeed Hiring Lab CSVs and the GPTs-are-GPTs labels when the dataset is built	Free, public
O*NET Resource Center (USDOL/ETA)	Downloads the O*NET 31.0 database when the dataset is built	Free, public

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale Deployment, Repl and Secrets (Replit)	Main live prototype at skilltide.replit.app: builds the static site and runs the Node.js API; the OpenAI key is kept in Replit Secrets	Replit Core (SDC sponsor code); Autoscale 2 vCPU / 4 GiB; region Asia

Service & Provider	Purpose in Project	Plan / Region
Cloudflare Workers with Static Assets (Cloudflare)	Second deployment at xmumhkt.arcueidmp.com: the site plus the API Worker; edge cache for AI-worded answers; Workers Logs	Workers Free; Cloudflare's global network
Cloudflare Workers Builds and DNS (Cloudflare)	Deploys every push to main; DNS for the custom domain	Free
GitHub repository and GitHub Actions (GitHub)	Version control; CI runs tests, lint and a static build on every push to main; a data-refresh workflow, paused during judging	Free; GitHub-hosted Ubuntu runners
Canva (Canva)	Pitch deck: designed in Canva, shared with a public view link and exported to PDF	A team member's Canva account

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-6-luna (reasoning effort "none")	OpenAI, through the Responses API	Pre-trained; called through the API, not fine-tuned. Used only when a student turns on AI wording	OpenAI Services Agreement and usage policies (proprietary)

No other AI model runs in SkillTide: CV skill recognition, scoring and the Guide's default answers use fixed rules and formulas.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 31.0 Database (USDOL/ETA)	onetcenter.org/database.html	Public	CC BY 4.0	N
Anthropic Economic Index (5 releases, Mar 2025 to Jun 2026) and robot exposure index	huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY (data)	N (aggregated)
GPTs are GPTs task exposure labels (Eloundou et al.)	github.com/openai/GPTs-are-GPTs	Public	MIT	N
Indeed Hiring Lab Job Postings Index and AI Tracker	github.com/hiring-lab	Public	CC BY 4.0	N
Hacker News "Who is hiring?" threads	hn.algolia.com/api	Public; only skill-mention counts are kept	Public API; post text not stored	N
UN SDG goals and targets	unstats.un.org/sdgs/UNSDGAPIV5	Public	UN terms of use; cited	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Microsoft Learn catalog (17 learning paths)	learn.microsoft.com/api/catalog	Public	Microsoft APIs Terms of Use (links and metadata)	N
SkillTide course checks (55 courses: hours and prices read from each provider's page)	github.com/ArcueidMP/skilltide, pipeline/curated/courses.json	Self-collected from public pages	CC BY 4.0 for our records; facts cited, no course content copied	N
MyCOL 2024/2025 (TalentCorp Malaysia)	talentcorp.com.my	Public	Public report; cited	N
New-occupation lists (China MOHRSS, TalentCorp, WEF)	Each item links to its source (pipeline/curated/emerging.json)	Public	Cited	N
Malaysian context figures (Hays, Bernama, The Star, DOSM, PwC, HRD Corp)	Each figure is shown with its link	Public	Cited	N
Synthetic students: sample CV (Kavin a/l Raj), 180-student cohort, 12 evaluation cases	public/samples/ and src/lib/ in the repository	Synthetic	Apache-2.0 (our code)	N (fictional)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code and claude.ai (Anthropic), Claude 5 models (mainly Opus 5.5)	Carrying out the team's plan: writing, testing and refactoring code (interface, API, scoring engine, data pipeline), drafting documents, the proposal and the pitch deck, building the demo-film tooling and setting up deployment, all under the team's direction and review	Major
Codex (OpenAI), gpt-6-astra	Carrying out parts of the team's plan, at the team's request: the CV PDF upload, the Cloudflare deployment set-up and the judge Q&A document	Moderate
Canva AI design generation (Canva), through Canva's connector in Claude	First layout of the pitch deck, generated from the team's outline; every slide's text was then rewritten, the images were replaced with screenshots of the app, and the team approved the result	Minor
Kokoro-82M through kokoro-onnx 0.6.1, run locally	Synthetic narration for an earlier cut of the demo film; the submitted film is narrated by a team member	Minor
Whisper base.en through FFmpeg's whisper filter, run locally	Checking that the film's narration is intelligible	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team planned SkillTide and made every product decision (users, scope, data rules, design, features); the AI tools carried out that plan, and we reviewed and tested the results. 106 automated tests (scoring engine, API, dataset) and 5 pipeline tests run with lint and a full build in GitHub Actions on every push to main, and a failed data refresh is never published. Course facts were read from each provider's page and stored with quotes and links. Every figure comes from the dataset or a cited source, and the team can explain every formula.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform and credits: Replit Core (redeemed with the SDC code), Autoscale deployment, Secrets	The GitHub repository is imported into the Repl @ArcueidMP/skilltide, which builds the site and runs the API. It is published as an Autoscale deployment (region Asia) at skilltide.replit.app, and the OpenAI key is kept in Replit Secrets.
RunPod	Not used	SkillTide trains and runs no model on a GPU, so it needed no GPU compute.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit hosts our main live prototype, which is how judges reach SkillTide, but the app does not depend on Replit-only services. It runs as one Node.js server that any Node host can run, and the same code is also deployed on Cloudflare Workers (xmumhkt.arcueidmp.com), so Replit could be replaced. We used no sponsor-provided GenAI API: the optional AI wording uses the team's own OpenAI key and is off by default, and without it the Guide answers with rules.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No accounts, no database: SkillTide's servers store no user data.

- **CV:** read in the browser with PDF.js; the file and full text are never uploaded.
- **Browser:** the profile (skills with the CV phrases behind them, goals, hours, budget, optional name), pathway progress and settings stay in local storage until the student presses Reset demo.
- **API:** requests carry skill ids and levels, SDG goals, weekly hours, budget, weights, finished-step ids and Guide questions, never the profile's name or CV text. Our code logs one line per request (path, status, duration), never the body or IP; the IP is used in memory only, to rate-limit AI requests.
- **OpenAI (only with AI wording on):** the typed question and SkillTide's computed facts are sent with storage disabled (store: false). On Cloudflare, the answer is cached for up to 7 days under a hash.
- Demo people are synthetic; no real CVs are used.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

- **Wrong or invented wording:** the model sees only SkillTide's facts. A reply that cites an unknown fact or contains a number not in its facts is discarded and the rule-based answer is shown. This cannot prove every sentence right, so AI wording is off by default and labelled on each answer.
- **One provider, limits and cost:** OpenAI errors, 12-second timeouts, refusals and our rate limits (per minute: 6 per client, 30 per server instance) fall back to the rules.
- **Data bias:** hiring demand comes from US, UK and Australian postings, and tasks from the US O*NET taxonomy; Malaysian signals come from MyCOL and cited reports. The app says so.
- **Estimates, not forecasts:** outlooks are formula-based Moderate and Fast scenarios.
- **Fairness:** scoring ignores name, age, gender and background; the check across name and graduation-year variants is narrow.
- **CV parsing:** fixed rules can miss skills, so students review and edit them.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

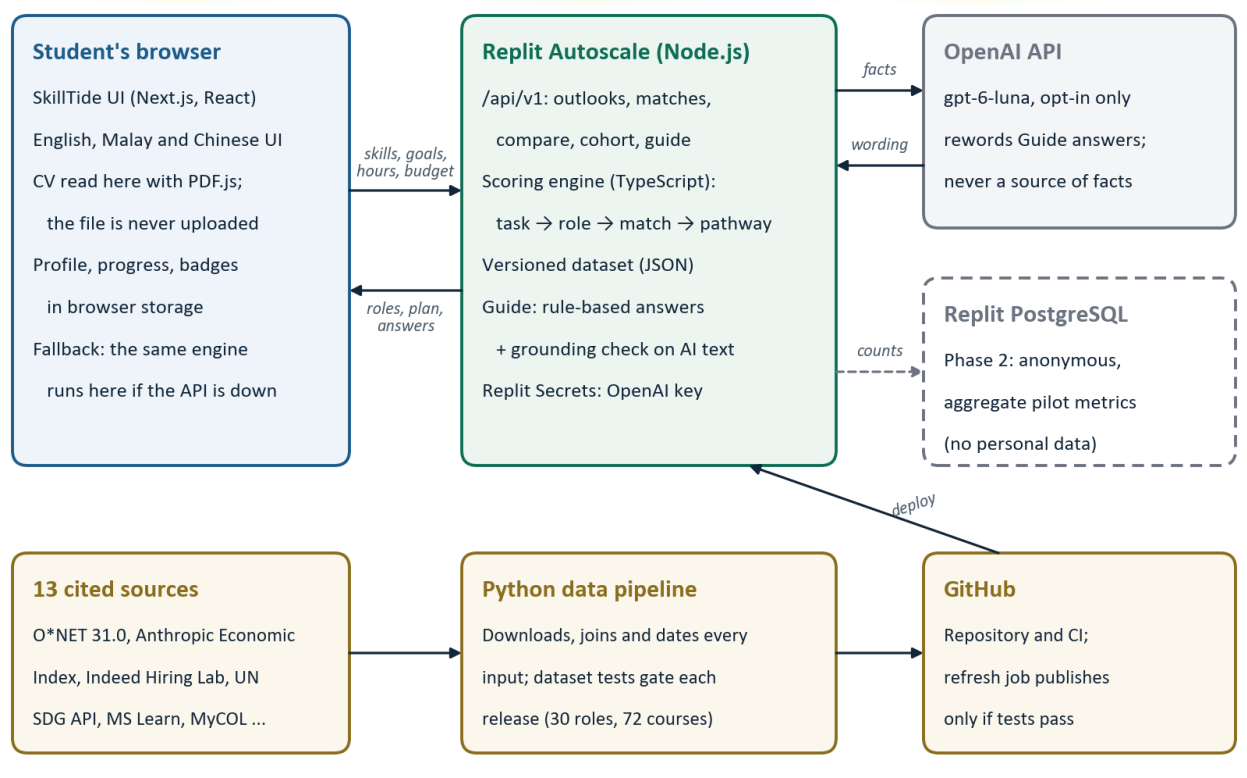
By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



HUANG BAONING, Team Leader

Date: 24 October 2026

Appendix A. System Block Diagram



The diagram from the preliminary proposal. The same API server also runs as a Cloudflare Worker at xmumhkt.arcueidmp.com. The dashed Replit PostgreSQL box is planned for Phase 2 and is not used yet.